

Faculty of Computers and Artificial Intelligence

Information Technology Department

Computer Networks

Network Design Project

FCAI Campus & Head Quarter (HQ)

Prepared by:

Kareem Abd EL Razik

Under supervision of:

Eng / Kareem Abd EL Razik

1. Introduction

In this project we designed a complete network for the Faculty of Computers and Artificial Intelligence (FCAI) and its Head Quarter (HQ). The goal was to connect all the labs, training centers, and departments in both sites using proper network design principles that we studied in the course.

The FCAI building has 2 floors with 4 labs/centers, and the HQ has 2 buildings with different departments. We used VLAN segmentation to separate the traffic between each lab and department, and we used DHCP so that all computers get their IP addresses automatically.

The two sites are connected together using a serial WAN link, and we configured default static routing between the two routers so they can communicate with each other.

The main technologies we used in this project:

- VLAN segmentation (VLANs 10, 20, 30, 40, 50, 60, 70, 80)
- Fixed-Length Subnet Masking (FLSM) starting from 192.168.1.0/29
- Inter-VLAN routing using Router-on-a-Stick method (sub-interfaces with dot1Q)
- DHCP Server for automatic IP assignment
- DNS Server for name resolution
- Default static routing between FCAI Router and HQ Router
- WAN serial link connecting both sites

2. Hardware Components Required

The following tables show all the hardware devices we used to build the network in Cisco Packet Tracer.

2.1 FCAI Site

Device	Type / Model	Quantity	Purpose
FCAI Router	Cisco 2911	1	Main router – inter-VLAN routing and WAN connection
Core Switch	Cisco 3560 (L3)	1	Connects everything at the core layer
Distribution Switches	Cisco 2960	2	One per floor, connects access switches
Access Switches	Cisco 2960	8	2 per lab/TC to connect the PCs
DHCP & DNS Server	PT Server	1	Gives IPs to all FCAI computers, also does DNS
Computers (PCs)	PC-PT	120	30 per lab/TC x 4 locations

2.2 HQ Site

Device	Type / Model	Quantity	Purpose
HQ Router	Cisco 2911	1	WAN connection + DHCP server for HQ + inter-VLAN
HQ Building Switch	Cisco 2960	1	Connects Project Manager and Team Leader
Collapsed Core Switch	Cisco 3560	1	Core of the Projects Building (two-tier design)
IT Access Switch	Cisco 2960	1	Connects IT1 and IT2
HR Access Switch	Cisco 2960	1	Connects HR1 and HR2
Computers (PCs)	PC-PT	6	IT x2, HR x2, Project Manager x1, Team Leader x1

3. Network Design Strategy

3.1 Three-Tier Hierarchical Design (FCAI)

For the FCAI campus we used the three-tier hierarchical model which is the standard design for large networks. This model divides the network into three layers:

- Core Layer: We used one Cisco 3560 Layer-3 switch as the core. It connects the router to both distribution switches and the data center.
- Distribution Layer: We have two distribution switches, one for floor 1 and one for floor 2. They connect the access layer switches to the core.
- Access Layer: We have 8 access switches in total (2 per lab/training center). The PCs connect directly to these switches.

3.2 Two-Tier Design (HQ Projects Building)

For the HQ Projects Building we used the two-tier collapsed core design because the building is small and has only 4 computers (IT1, IT2, HR1, HR2). In this design we combine the core and distribution into one switch called the Collapsed Core Switch. This saves money and reduces complexity.

3.3 HQ Main Building

The HQ main building only has 2 computers: the Project Manager (VLAN 70) and the Team Leader (VLAN 80). We connected them using one access switch that connects directly to the HQ Router. The HQ Router does the inter-VLAN routing for them using sub-interfaces.

3.4 WAN Connection Between Sites

We connected the FCAI Router to the HQ Router using a serial WAN link. The WAN network address is 192.168.1.64/30 which gives us 2 usable IPs: 192.168.1.65 for the FCAI Router and 192.168.1.66 for the HQ Router. We chose default static routing because the topology is simple and there is only one path between the two sites, so we do not need a dynamic routing protocol.

4. VLAN and IP Network Design

4.1 Subnetting Using FLSM

The base network given in the project is 192.168.1.0/29. A /29 subnet has 8 addresses (6 usable hosts). We need 8 VLANs so we divided the address space into 8 subnets of the same size (/29). We also added a /30 subnet for the WAN link between the two routers.

VLAN	Name	Network Address	Subnet Mask	Gateway	Host Range	Broadcast
10	Lab3	192.168.1.0	255.255.255.248	192.168.1.1	192.168.1.2 – .6	192.168.1.7
20	Training Center	192.168.1.8	255.255.255.248	192.168.1.9	192.168.1.10 – .14	192.168.1.15
30	Lab1	192.168.1.16	255.255.255.248	192.168.1.17	192.168.1.18 – .22	192.168.1.23
40	Lab2	192.168.1.24	255.255.255.248	192.168.1.25	192.168.1.26 – .30	192.168.1.31
50	IT Department	192.168.1.32	255.255.255.248	192.168.1.33	192.168.1.34 – .38	192.168.1.39
60	HR Department	192.168.1.40	255.255.255.248	192.168.1.41	192.168.1.42 – .46	192.168.1.47
70	Project Manager	192.168.1.48	255.255.255.248	192.168.1.49	192.168.1.50 – .54	192.168.1.55
80	Team Leader	192.168.1.56	255.255.255.248	192.168.1.57	192.168.1.58 – .62	192.168.1.63
WAN	FCAI ↔ HQ	192.168.1.64	255.255.255.252	—	192.168.1.65 – .66	192.168.1.67

4.2 DHCP Configuration

We configured DHCP in two ways depending on the location:

- FCAI Floors 1 and 2: We use a dedicated DHCP Server located in the Data Center (IP: 192.168.1.2, VLAN 10). The FCAI Router uses the ip helper-address command on each sub-interface to forward DHCP requests from VLANs 10, 20, 30, and 40 to this server.
- HQ Site: The HQ Router itself acts as a DHCP server. We created 4 DHCP pools on the HQ Router, one for each VLAN (50, 60, 70, 80).

5. Network Topology Diagram

The following figure shows the full network topology as we built it in Cisco Packet Tracer. The diagram includes all devices, connections, and VLAN labels.

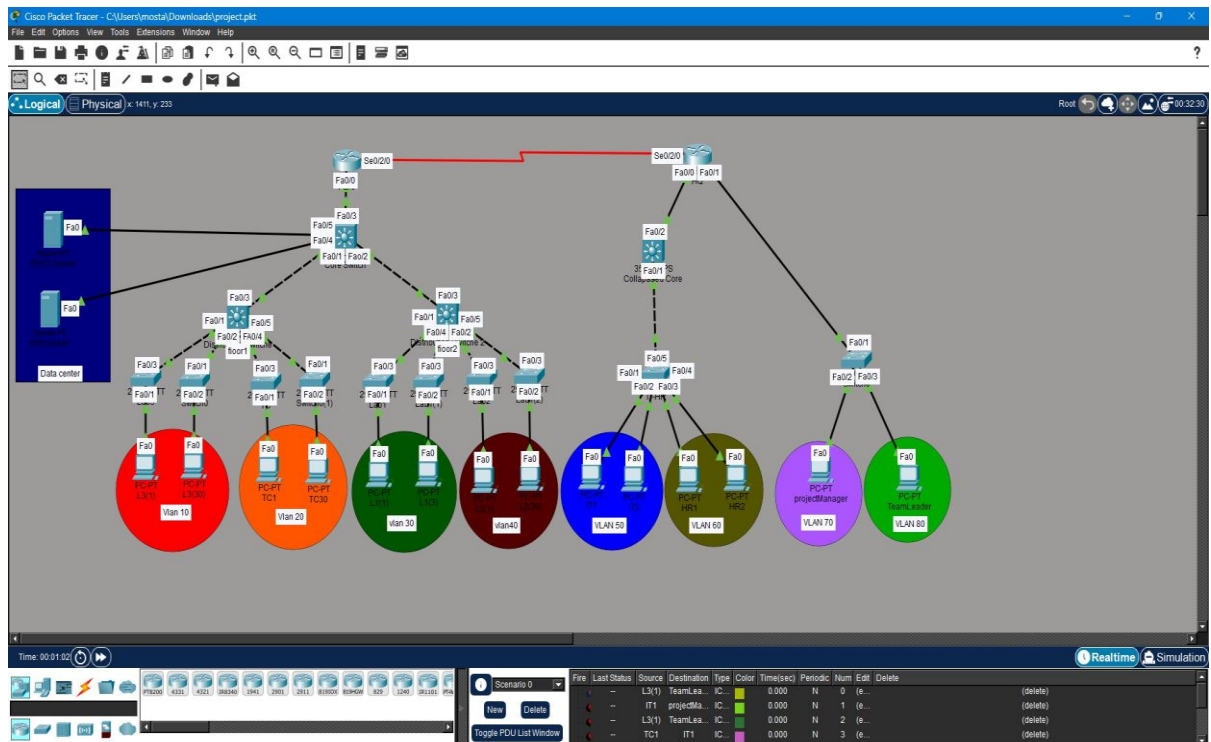


Figure 1: FCAI & HQ Network Topology – Cisco Packet Tracer

6. Network Diagram Explained

FCAI Campus Side

The FCAI Router is the main device on this side. It connects to the Core Switch using a trunk link, and also connects to the HQ Router using the serial WAN interface. All inter-VLAN routing for VLANs 10, 20, 30, and 40 is done by the router using sub-interfaces (Router-on-a-Stick).

The Data Center is connected to the Core Switch and contains the DHCP Server (192.168.1.2) and the DNS Server. Both servers are on the same machine in Packet Tracer. The DHCP Server is in VLAN 10 and it serves all four VLANs through the ip helper-address configured on the router.

The Core Switch connects to two Distribution Switches, one for floor 1 and one for floor 2. Floor 1 has Lab3 (VLAN 10) and the Training Center (VLAN 20). Floor 2 has Lab1 (VLAN 30) and Lab2 (VLAN 40). Each lab has 2 access switches, each one handles 15 computers.

HQ Side

The HQ Router is the main device on the HQ side. It handles the WAN connection, does inter-VLAN routing for all HQ VLANs (50, 60, 70, 80), and also works as a DHCP server for all HQ computers.

The HQ main building has a simple access switch connected directly to the HQ Router. The Project Manager computer is in VLAN 70 and the Team Leader is in VLAN 80. The router uses sub-interfaces with dot1Q for the inter-VLAN routing between them.

The Projects Building uses a Collapsed Core Switch connected to the HQ Router. Under it there are 2 access switches, one for IT (VLAN 50) with 2 computers, and one for HR (VLAN 60) with 2 computers.

7. Network Configuration

7.1 FCAI Router Configuration

```
hostname FCAI-Router
!
interface Serial0/2/0
 ip address 192.168.1.65 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 no ip address
 no shutdown
!
interface FastEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.1 255.255.255.248
 ip helper-address 192.168.1.2
!
interface FastEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.1.9 255.255.255.248
 ip helper-address 192.168.1.2
!
interface FastEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.1.17 255.255.255.248
 ip helper-address 192.168.1.2
!
interface FastEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.1.25 255.255.255.248
 ip helper-address 192.168.1.2
!
ip route 0.0.0.0 0.0.0.0 192.168.1.66
ip name-server 192.168.1.2
```

7.2 HQ Router Configuration

```
hostname HQ-Router
!
interface Serial0/2/0
 ip address 192.168.1.66 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 no ip address
 no shutdown
!
interface FastEthernet0/0.70
 encapsulation dot1Q 70
 ip address 192.168.1.49 255.255.255.248
!
interface FastEthernet0/0.80
 encapsulation dot1Q 80
```

```

ip address 192.168.1.57 255.255.255.248
!
interface FastEthernet0/1
no ip address
no shutdown
!
interface FastEthernet0/1.50
encapsulation dot1Q 50
ip address 192.168.1.33 255.255.255.248
!
interface FastEthernet0/1.60
encapsulation dot1Q 60
ip address 192.168.1.41 255.255.255.248
!
ip dhcp excluded-address 192.168.1.33
ip dhcp excluded-address 192.168.1.41
ip dhcp excluded-address 192.168.1.49
ip dhcp excluded-address 192.168.1.57
!
ip dhcp pool VLAN50-IT
network 192.168.1.32 255.255.255.248
default-router 192.168.1.33
dns-server 192.168.1.2
!
ip dhcp pool VLAN60-HR
network 192.168.1.40 255.255.255.248
default-router 192.168.1.41
dns-server 192.168.1.2
!
ip dhcp pool VLAN70-PM
network 192.168.1.48 255.255.255.248
default-router 192.168.1.49
dns-server 192.168.1.2
!
ip dhcp pool VLAN80-TL
network 192.168.1.56 255.255.255.248
default-router 192.168.1.57
dns-server 192.168.1.2
!
ip route 0.0.0.0 0.0.0.0 192.168.1.65

```

7.3 DHCP Server Settings (Packet Tracer GUI)

The DHCP Server is in the Data Center. We gave it a static IP of 192.168.1.2 (in VLAN 10). Then we opened the Server GUI > Services > DHCP and created 4 pools as follows:

Pool Name	Network	Subnet Mask	Default GW	Start IP	Max Users
VLAN10-Lab3	192.168.1.0	255.255.255.248	192.168.1.1	192.168.1.3	4
VLAN20-TC	192.168.1.8	255.255.255.248	192.168.1.9	192.168.1.10	5
VLAN30-Lab1	192.168.1.16	255.255.255.248	192.168.1.17	192.168.1.18	5

Pool Name	Network	Subnet Mask	Default GW	Start IP	Max Users
VLAN40-Lab2	192.168.1.24	255.255.255.248	192.168.1.25	192.168.1.26	5

Note: For VLAN 10, the Start IP is 192.168.1.3 (not .2) because 192.168.1.2 is already the static IP of the DHCP Server itself.

7.4 Core Switch (FCAI)

```
hostname FCAI-CoreSwitch
vlan 10
    name Lab3
vlan 20
    name TrainingCenter
vlan 30
    name Lab1
vlan 40
    name Lab2
!
interface GigabitEthernet1/0/1
    switchport mode trunk
    switchport trunk allowed vlan 10,20,30,40
!
interface FastEthernet0/1
    switchport mode trunk
    switchport trunk allowed vlan 10,20
!
interface FastEthernet0/2
    switchport mode trunk
    switchport trunk allowed vlan 30,40
!
interface FastEthernet0/3
    switchport mode access
    switchport access vlan 10
```

7.5 Distribution Switch – Floor 1

```
hostname Floor1-DistSwitch
vlan 10
    name Lab3
vlan 20
    name TrainingCenter
interface FastEthernet0/1
    switchport mode trunk
    switchport trunk allowed vlan 10,20
interface FastEthernet0/2
    switchport mode trunk
    switchport trunk allowed vlan 10
interface FastEthernet0/3
    switchport mode trunk
    switchport trunk allowed vlan 10
interface FastEthernet0/4
    switchport mode trunk
    switchport trunk allowed vlan 20
interface FastEthernet0/5
    switchport mode trunk
```



```
switchport trunk allowed vlan 20
```

7.6 Distribution Switch – Floor 2

```
hostname Floor2-DistSwitch
vlan 30
  name Lab1
vlan 40
  name Lab2
interface FastEthernet0/1
  switchport mode trunk
  switchport trunk allowed vlan 30,40
interface FastEthernet0/2
  switchport mode trunk
  switchport trunk allowed vlan 30
interface FastEthernet0/3
  switchport mode trunk
  switchport trunk allowed vlan 30
interface FastEthernet0/4
  switchport mode trunk
  switchport trunk allowed vlan 40
interface FastEthernet0/5
  switchport mode trunk
  switchport trunk allowed vlan 40
```

7.7 Access Switches – Lab3 (x2), Training Center (x2), Lab1 (x2), Lab2 (x2)

All access switches follow the same configuration pattern. Below is an example for Lab3. The same steps apply for TC (vlan 20), Lab1 (vlan 30) and Lab2 (vlan 40):

```
hostname SW-Lab3-1
vlan 10
  name Lab3
interface FastEthernet0/1
  switchport mode trunk
  switchport trunk allowed vlan 10
interface range FastEthernet0/2 - 16
  switchport mode access
  switchport access vlan 10
  spanning-tree portfast
```

7.8 HQ Building Switch

```
hostname HQ-Switch
vlan 70
  name ProjectManager
vlan 80
  name TeamLeader
interface FastEthernet0/1
  switchport mode trunk
  switchport trunk allowed vlan 70,80
interface FastEthernet0/2
  switchport mode access
  switchport access vlan 70
  spanning-tree portfast
interface FastEthernet0/3
  switchport mode access
  switchport access vlan 80
  spanning-tree portfast
```

7.9 Projects Building – Collapsed Core Switch

```
hostname Projects-CollapsedCore
vlan 50
    name IT
vlan 60
    name HR
interface FastEthernet0/1
    switchport mode trunk
    switchport trunk allowed vlan 50,60
interface FastEthernet0/2
    switchport mode trunk
    switchport trunk allowed vlan 50
interface FastEthernet0/3
    switchport mode trunk
    switchport trunk allowed vlan 60
```

7.10 IT & HR Access Switches

```
hostname SW-IT
vlan 50
    name IT
interface FastEthernet0/1
    switchport mode trunk
    switchport trunk allowed vlan 50
interface FastEthernet0/2
    switchport mode access
    switchport access vlan 50
    spanning-tree portfast
interface FastEthernet0/3
    switchport mode access
    switchport access vlan 50
    spanning-tree portfast
!
hostname SW-HR
vlan 60
    name HR
interface FastEthernet0/1
    switchport mode trunk
    switchport trunk allowed vlan 60
interface FastEthernet0/2
    switchport mode access
    switchport access vlan 60
    spanning-tree portfast
interface FastEthernet0/3
    switchport mode access
    switchport access vlan 60
    spanning-tree portfast
```

7.11 DNS Server Configuration

We enabled the DNS service on the server (Server > Services > DNS > ON) and added A records for the different parts of the network:

Record Name	IP Address	Meaning
fcai.edu	192.168.1.1	FCAI Router gateway
dns.fcai.edu	192.168.1.2	DNS/DHCP Server

Record Name	IP Address	Meaning
lab3.fcai.edu	192.168.1.1	VLAN 10 gateway
tc.fcai.edu	192.168.1.9	VLAN 20 gateway
lab1.fcai.edu	192.168.1.17	VLAN 30 gateway
lab2.fcai.edu	192.168.1.25	VLAN 40 gateway
hq.net	192.168.1.49	HQ Router VLAN 70 interface
pm.hq.net	192.168.1.50	Project Manager
tl.hq.net	192.168.1.58	Team Leader

8. Summary

In this project we successfully designed and configured a full network for FCAI and HQ using Cisco Packet Tracer. We covered all the requirements mentioned in the project sheet:

Requirement	What We Did	Done?
Use 192.168.1.0/29 network	Divided into 8 subnets /29 + /30 for WAN	✓
FLSM subnetting	Each VLAN gets one /29 subnet (6 usable IPs)	✓
Floors 1 & 2 get IPs from DHCP Server	Used ip helper-address on FCAI Router	✓
HQ devices get IPs from HQ Router	Configured 4 DHCP pools directly on HQ Router	✓
Enable DNS Server	DNS server at 192.168.1.2 with all A records	✓
Routing between FCAI and HQ	Default static routing on both routers	✓
Inter-VLAN for HQ Building (VLAN 70 & 80)	Sub-interfaces + dot1Q trunk on HQ Router	✓
Three-tier (FCAI) / Two-tier (HQ Projects)	Implemented exactly as required	✓

All devices were configured and tested in Packet Tracer. All PCs received their IPs from DHCP, all VLANs can communicate through the routers, and both sites can ping each other through the WAN link.